

Project Summary

Team name	SENTRY	
Project title	Low-Cost Bridge Vibration Anomaly Detection System Robust to Temperature and Humidity	
Role	Name	Affiliation
Participants	Kim Changhoon	CheongShim International Academy
	[name withheld]	CheongShim International Academy
Advisor	[name withheld]	CheongShim International Academy
Project summary		
1. Project topic We developed a low-cost first-stage bridge screening system that records vibration and the environment together with a MEMS accelerometer and a temperature/humidity sensor, compensates for normal variation explained by temperature and humidity, alerts only on the remaining change as anomaly candidates, and also shows which vibration characteristic departed. 2. Description of the project (proposed solution and its rationale) Class 3 small and medium bridges, 64.5% of all bridges in Korea, lack a low-cost means of continuous monitoring to fill the gap between precision inspections. This project collects vibration and temperature/humidity with an ADXL355Z, a DHT22 and a Raspberry Pi 5 (core parts about ₩190,000); it removes the part explained by a temperature/humidity regression from eight log/ratio-normalized vibration features, feeds the residuals to an Autoencoder trained only on normal data, flags an anomaly candidate when the reconstruction error exceeds the threshold, and reports the contribution of each feature. The design rests on the facts that natural frequencies change with temperature and humidity even without damage and that damage data cannot be obtained from bridges in service. 3. Differences from existing work (solutions) Existing low-cost MEMS studies do not address environmental compensation, fiber-optic (BOCDA) precision measurement requires dedicated infrastructure, and unsupervised studies on public data mostly stop at outputting an anomaly score. This project integrates measurement, environmental compensation, anomaly judgment and explanation of causes into a single low-cost device and validated it directly, from model experiments to real pedestrian overpasses. 4. Results In the 27-condition environment × mass × bolt experiment on the aluminum model, the compensated model detected all 840 damage windows (recall 100%) with false alarms on only 4 of 90 normal windows (4.4%), and its contributions distinguished bolt loosening from mass change. On public openLAB data, once the frequency channel was combined, temperature compensation reduced the autumn false-alarm rate from 26% to 0-3%. On two real overpasses, mode peaks were tracked within 1%; with a 200 kg load the frequency fell by 0.6% and returned after removal, and alarms sounded only in the loaded state, not in the unloaded state.		